

Actuator Optimal Design

Lecture 2 : SPMSM Modeling

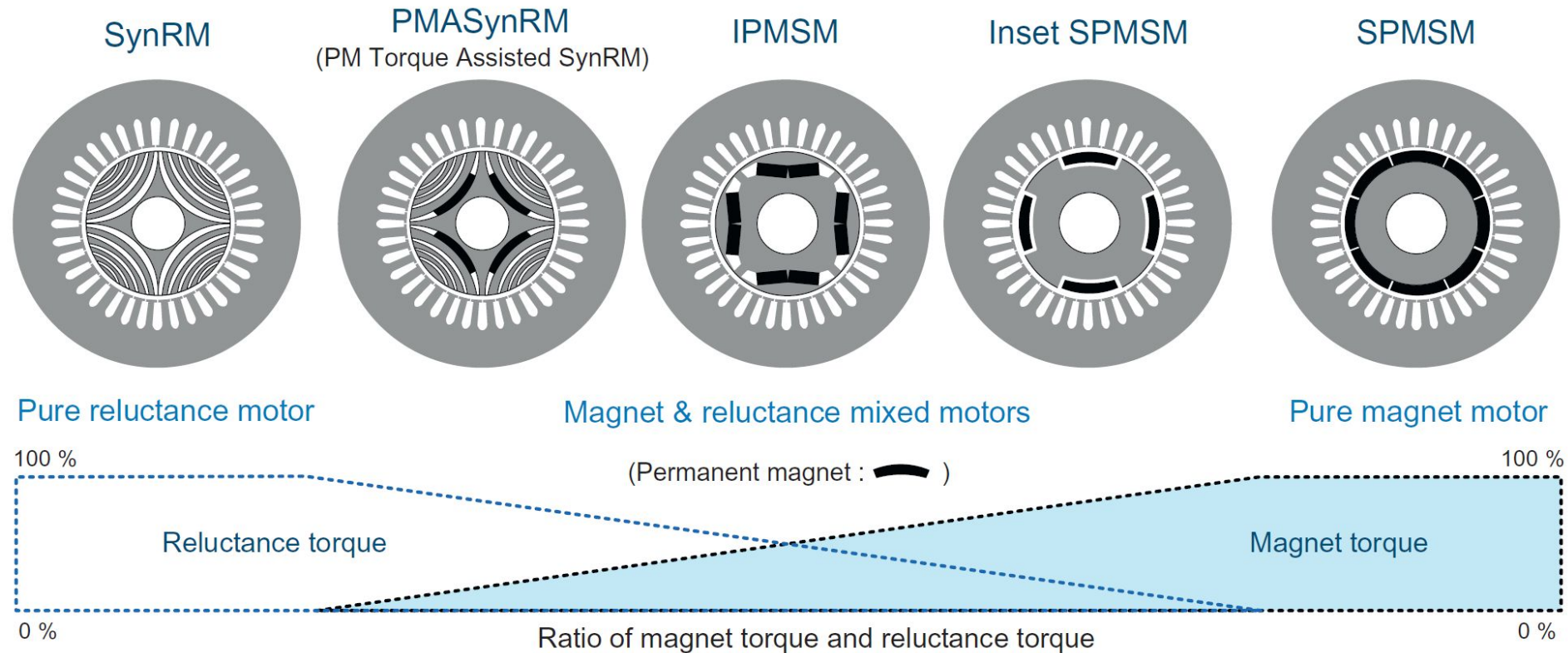
2026/03/09

KAIST DRCD lab 송태규 박사과정

Types of Motors

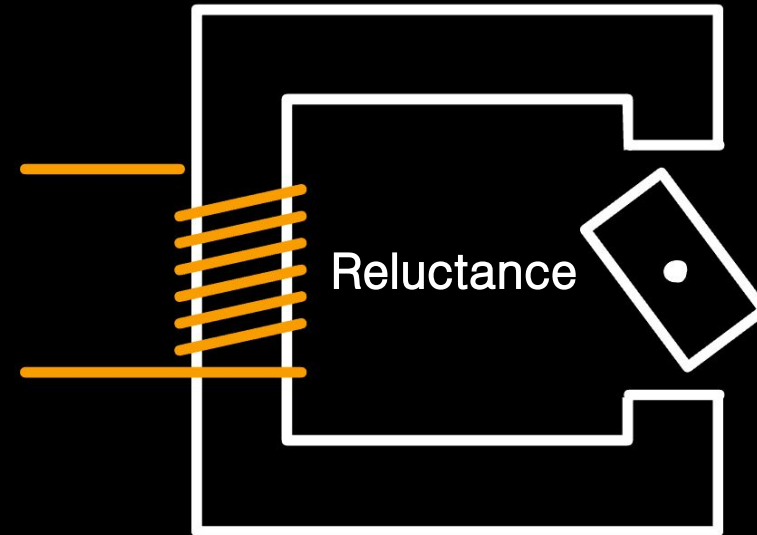
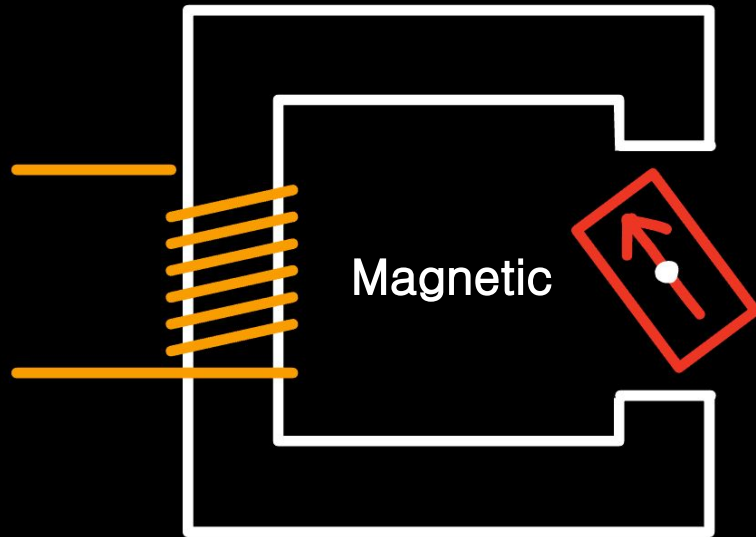
Types of Motors

- SPMSM: **S**urface **P**ermanent **M**agnet **S**ynchronous **M**otor
- IPMSM: **I**nterior **P**ermanent **M**agnet **S**ynchronous **M**otor
- SynRM: **S**ynchronous **R**eluctance **M**otor



Types of Motors

- Magnetic Torque & Reluctance Torque

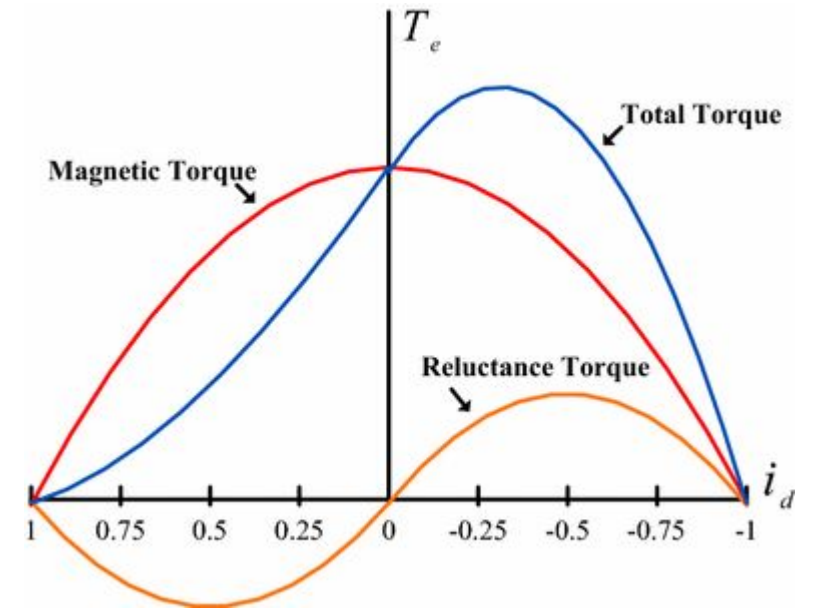
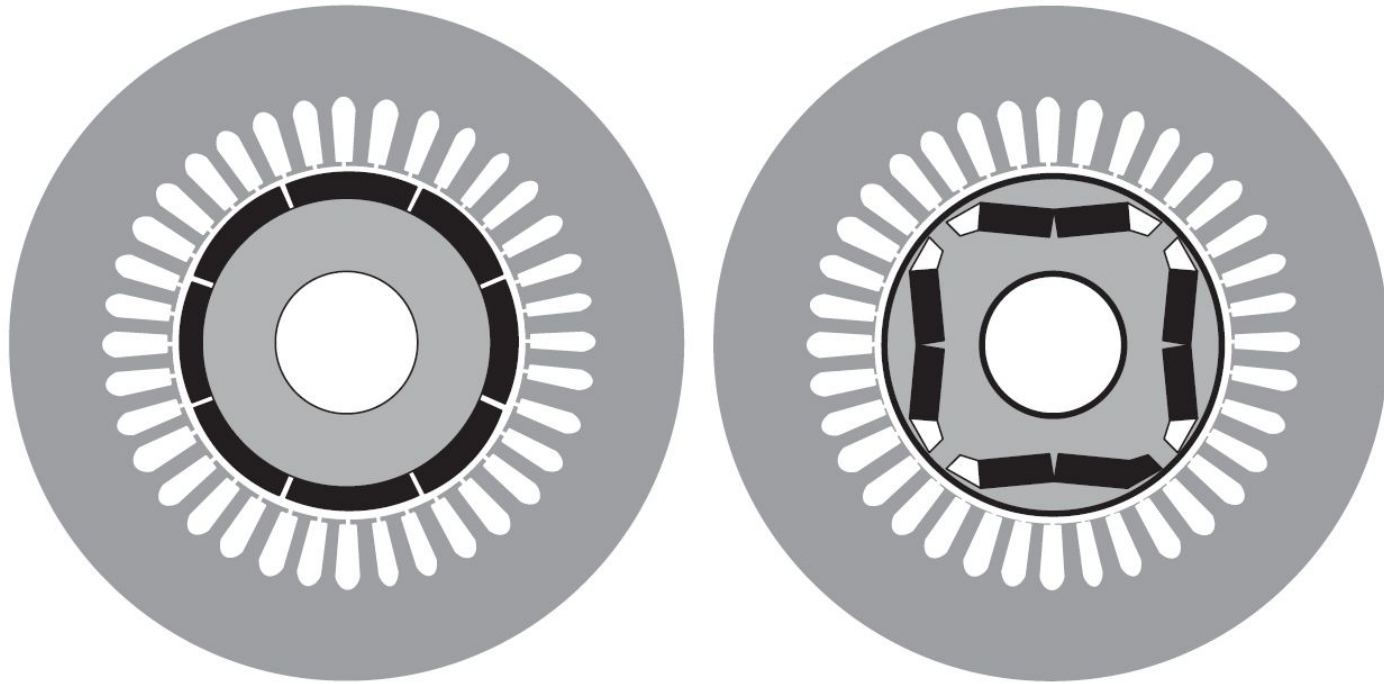


Types of Motors

- [Quiz]

Why No IPMSMs in Robotics?

(A)

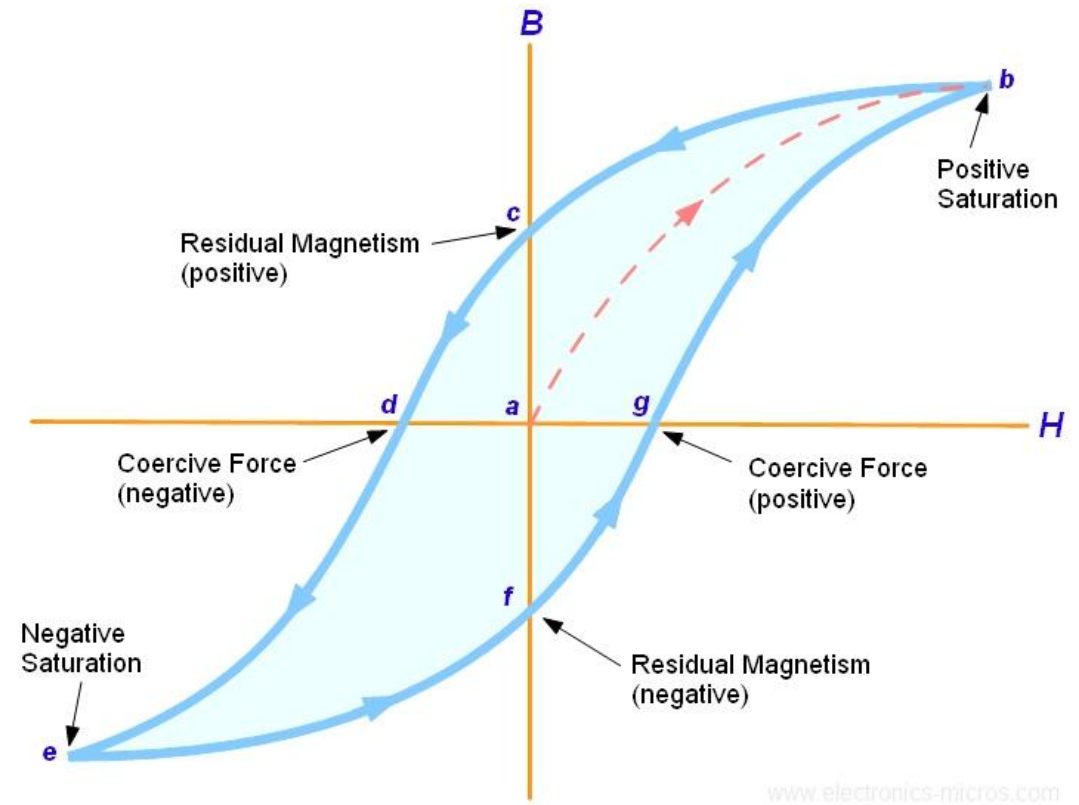
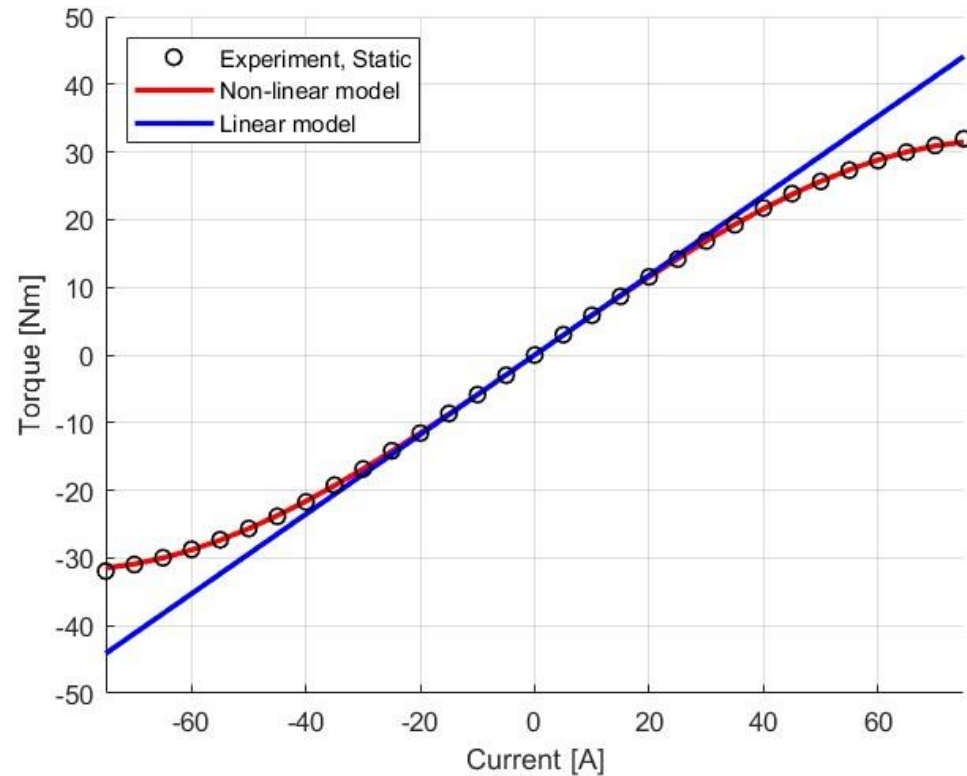


SPMSM Modeling

SPMSM Modeling : Basics

- Torque - Current model :

$$\tau = K_T i_{qs} + c_{sat} i_{qs}^3 \text{ (ex)}$$



SPMSM Modeling : Basics

- Brushed DC motor model :

$$V = iR + K_b\omega + L\frac{di}{dt} \text{ (neglected)}$$

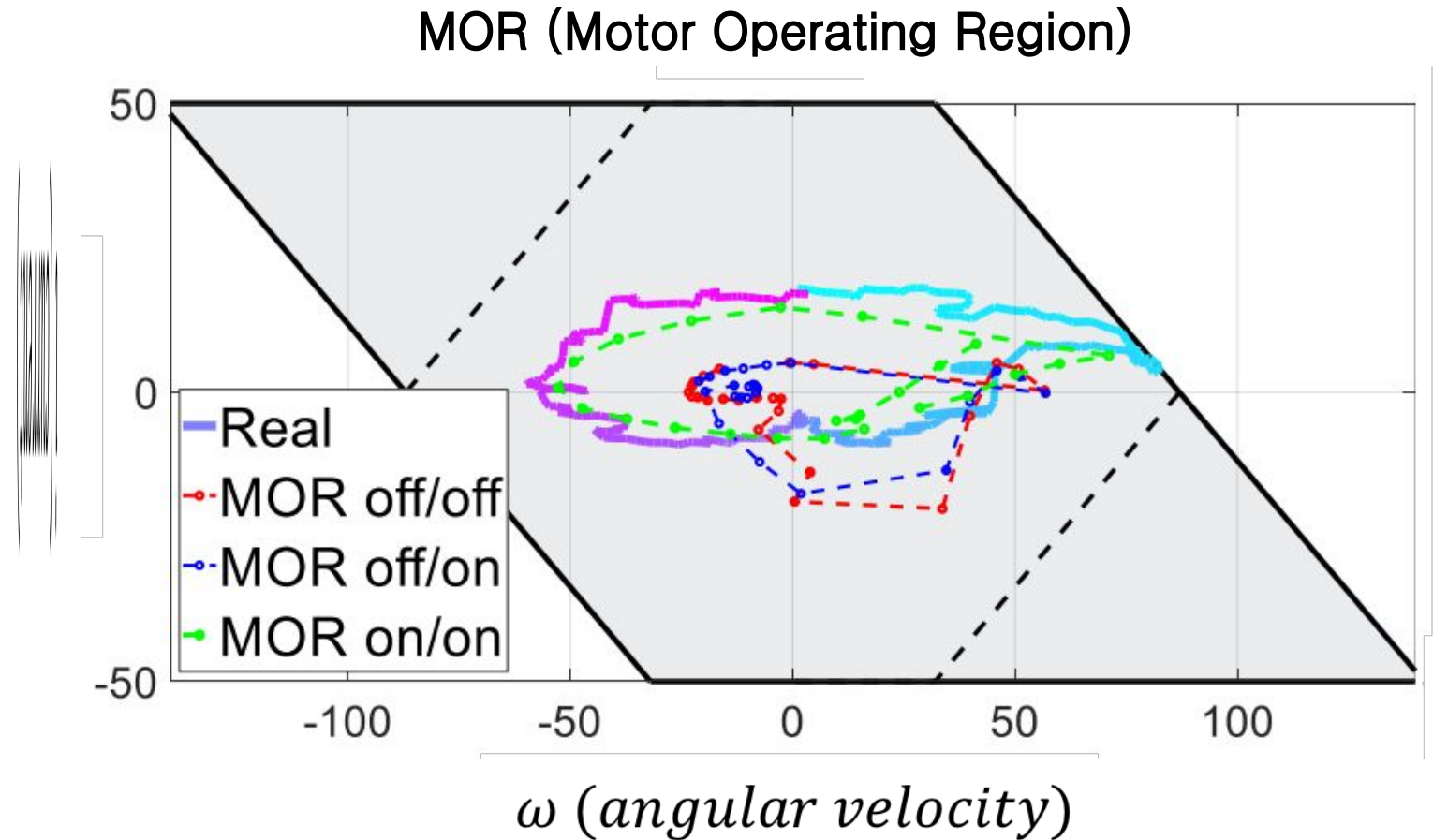
$$1) -V_{dc} < V < +V_{dc}$$

$$2) -i_{max} < i < +i_{max}$$

iR : Low speed dominant

$K_b\omega$: High speed dominant

$L\frac{di}{dt}$: Neglectable(usually)

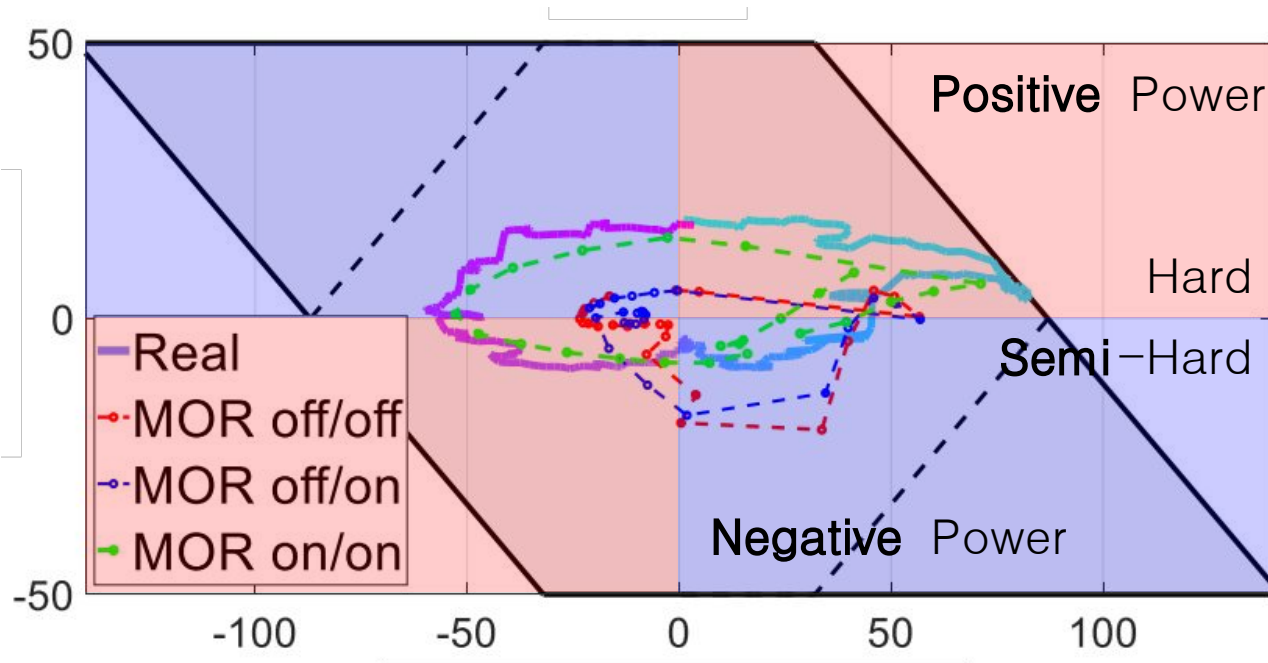


SPMSM Modeling : Advanced

- Brushed DC motor model :

$$V = iR + K_b\omega + L\frac{di}{dt} \text{ (neglected)}$$

- 1) $-V_{dc} < V < +V_{dc}$: Voltage limit const. (Hard) due to Battery (V_{dc}).
- 2) $-i_{max} < i < +i_{max}$: Current limit const. (Soft) due to Thermal limit ($P_{heat} = i^2 R$) or Core saturation.



- Semi-Hard?
Real battery is **NOT** Ideal Voltage source.
Internal resistance exists (IR drop).

SPMSM Modeling : Advanced

- SPMSM(Surface Permanent Magnet Synchronous Motor) model :

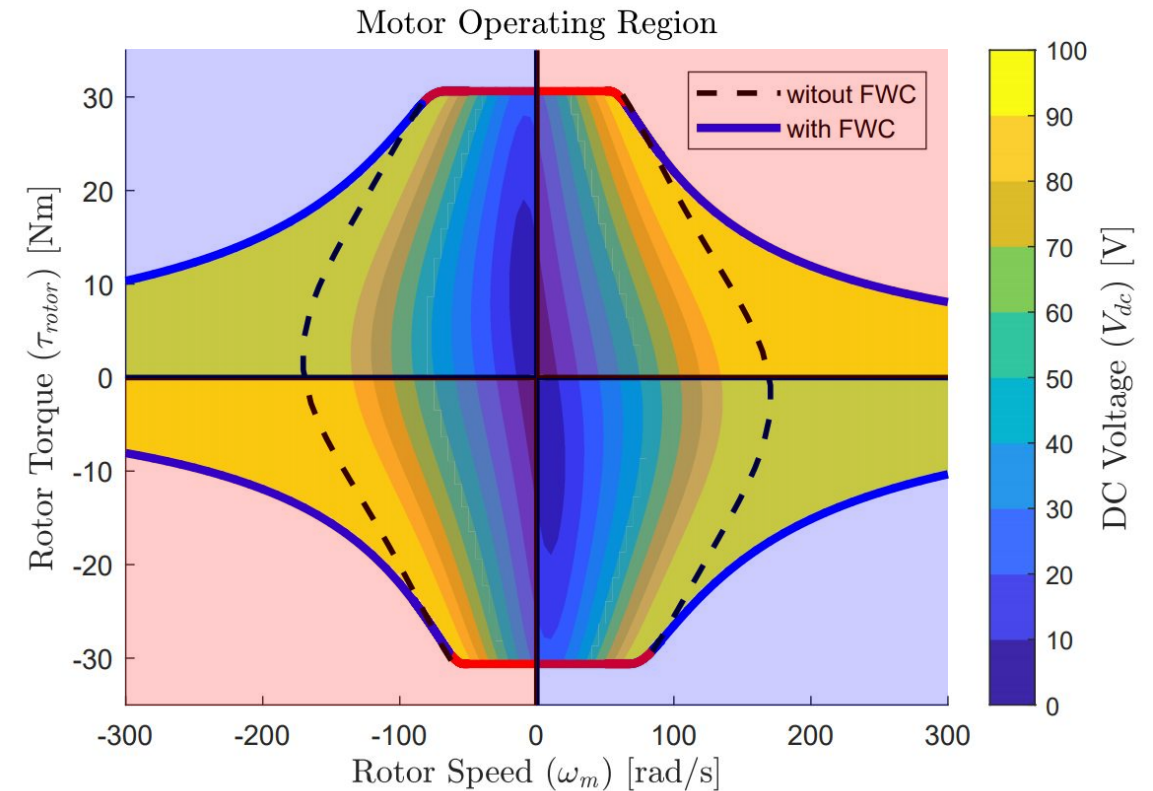
$$v_{ds} = R_s i_{ds} - \omega_e L_s i_{qs} + L_s \frac{di_{ds}}{dt}$$

$$v_{qs} = R_s i_{qs} + \omega_e L_s i_{ds} + \omega_e \varphi_f + L_s \frac{di_{qs}}{dt}$$

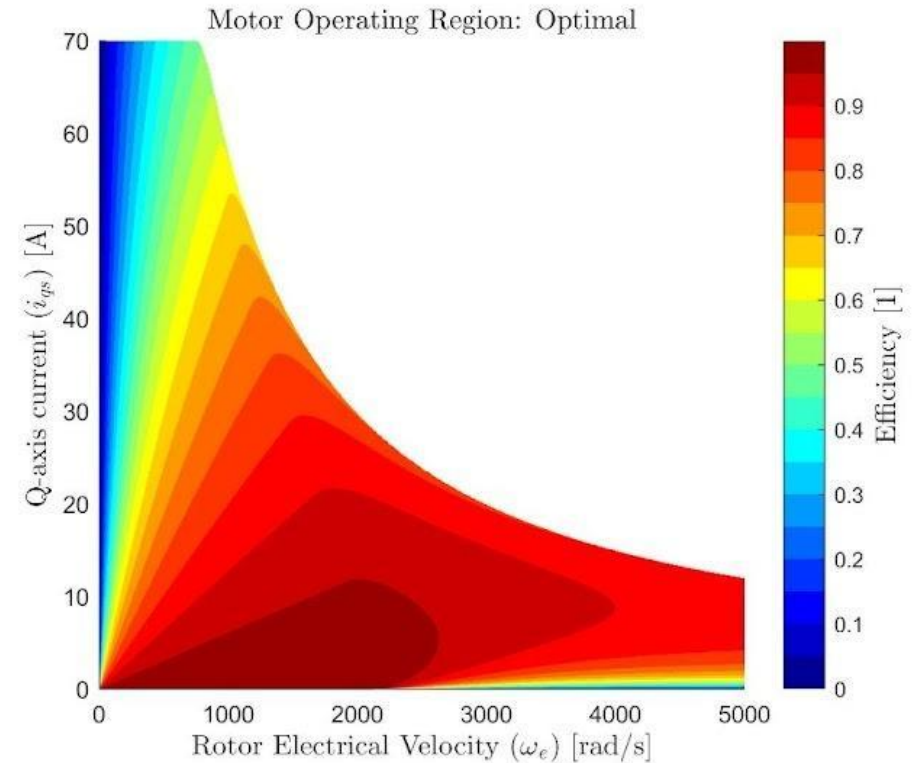
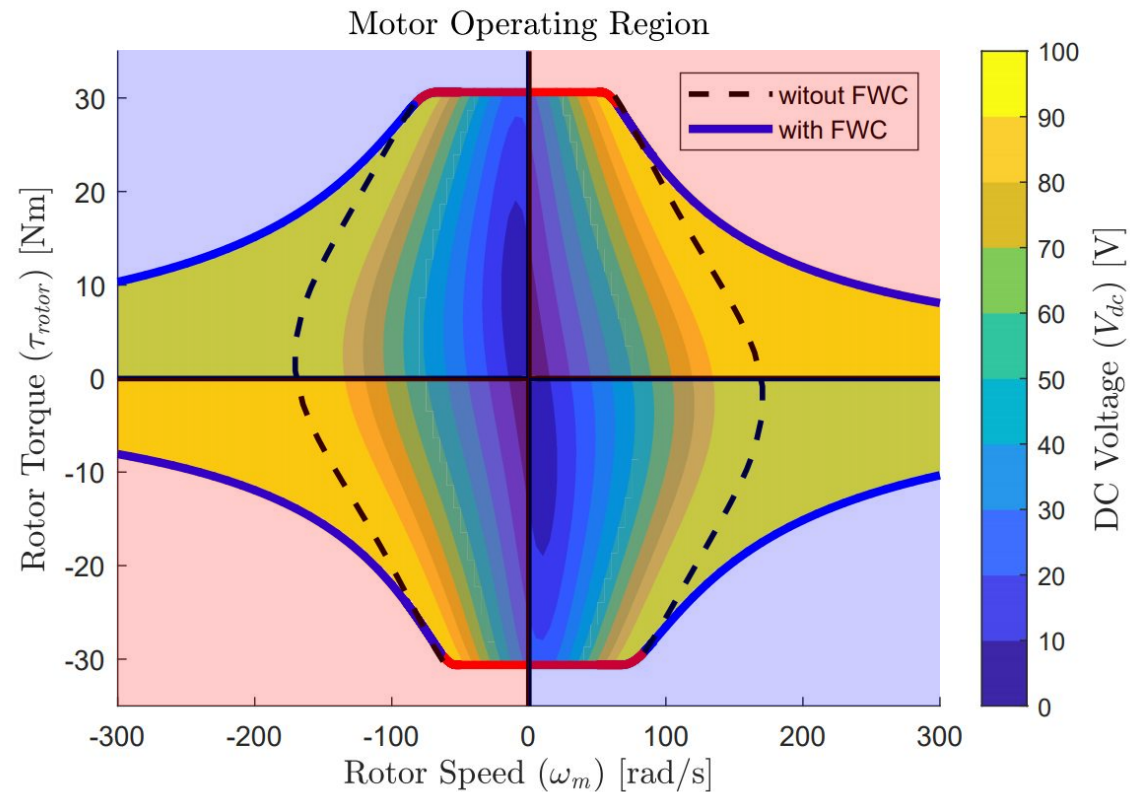
$$\tau_{rotor} = \frac{3P}{2} \varphi_f i_{qs}, \quad \omega_e = \frac{P}{2} \omega_m$$

$$1) \sqrt{v_{ds}^2 + v_{qs}^2} < V_{s,max}, \quad V_{s,max} = \frac{V_{dc}}{\sqrt{3}} \text{ (ideal PWM)}$$

$$2) \sqrt{i_{ds}^2 + i_{qs}^2} < i_{s,max}$$



- D-Q coupling (if 0, same model with Brushed DC motor)



How can we draw these?

By voltage equations.